

Disruptive Technologies-1
[23ECH-102]
Project Report
On
DriveMatch -
Suggest the car to the user according to his specifications .

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2023

1.Introduction :

Choosing the right car in India can be tough with so many options available. This project is about making that choice easier. We're creating a special system that suggests cars based on what people want in India.

In India, everyone has different needs when it comes to cars. Some want fuel-efficient cars, others need ones that handle specific roads well. Our system aims to help by using smart technology to find cars that match what people are looking for.

We'll be using data and computer smarts to make this happen.

This means we'll collect information about lots of cars available in India. Then, our system will use this info to suggest cars that fit each person's budget, preferences, and needs.

India is a big place with different regions and different preferences. Our system will consider all these differences. Whether it's about roads, the kind of fuel available, or what people like in different parts of India, our system will try to suggest cars that suit each person perfectly.

This report will explain how we collected data, how our smart system works, and what we found out from using it. We'll talk about why suggesting the right cars in India matters and how we can make this system even better in the future.

This project is all about making it easier for people in India to find the perfect car that suits them best.

2. Brief overview of the project :

This project revolves around creating a user-friendly system that recommends cars tailored to individual preferences in India. By leveraging data analytics and machine learning, the system aims to simplify the daunting task of choosing the right car amidst the diverse automotive market in the country.

The project encompasses the collection of extensive car-related data specific to India, including factors like budget, fuel efficiency, road conditions, regional preferences, and more. Through a meticulously designed recommendation algorithm, the system matches user-inputted criteria with the available car options, providing personalized suggestions.

The core objective is to empower users by presenting them with a curated selection of cars that align with their unique needs, preferences, and constraints. The system takes into account India's diverse landscape, considering regional variations and preferences to ensure suitability across different areas.

This brief overview highlights the project's focus on creating an accessible and efficient platform that simplifies the car selection process for individuals in India, ultimately aiding them in making informed and personalized choices.

3. Project Description

User Input:

Begin by accessing the AutoMatch platform.

Input your specific requirements, including budget constraints, preferred fuel efficiency, desired features (e.g., safety features, entertainment options), and any other relevant criteria.

Data Analysis:

AutoMatch utilizes advanced algorithms to analyze a comprehensive database of cars, taking into account various factors such as price, fuel economy, performance, and features.

Matching Algorithm:

The system employs a sophisticated matching algorithm to identify cars that closely align with your specified preferences. This algorithm considers not only explicit criteria but also factors in implicit preferences based on user patterns.

Personalized Recommendations:

Receive a personalized list of recommended cars that best fit your criteria. Each suggestion includes detailed information about

the car, ensuring you have a comprehensive understanding of why it's a good match.

Comparative Analysis:

Explore side-by-side comparisons of the suggested cars, highlighting key differences and similarities. This feature enables you to make well-informed decisions based on a thorough understanding of each option.

Feedback and Refinement:

Provide feedback on the recommendations to further refine the system's understanding of your preferences. This iterative process ensures that subsequent suggestions become increasingly tailored to your unique needs.

Access to Additional Information:

Access detailed specifications, user reviews, and expert opinions for each suggested car, empowering you with comprehensive information to make a confident decision.

Decision-Making Support:

AutoMatch goes beyond mere recommendations, offering insights and advice to assist you in making the final decision. This support includes factors like long-term cost of ownership, reliability ratings, and resale value.

User Interaction:

Enjoy a user-friendly interface that allows for easy navigation and interaction throughout the entire process.

Seamless Car Selection:

With the insights gained from the AutoMatch system, confidently select the car that best meets your needs, preferences, and budget.

Feel free to provide your specifications, and let's start the process of finding your ideal car!

4.Project Code

```
[  
import pandas as pd  
df=pd.read_csv("/content/CARS DATA 2.csv")  
df
```

```
[ ]
# Drop some unwanted columns in DATASET

#syntax for drop any Row/Column
#df.drop("Name of row/column", inplace = True, axis = 0 or 1)    0 for row , 1
for column
df.drop("Price", inplace = True , axis=1)
df
```

```
[ ]
z=['Levy','Cylinders','Doors']

for i in z:
    df.drop( i,inplace = True , axis = 1)
```

```
[ ]
en=str(input("enter engine type : "))
i=0
d_1=[]
while i<8244:
    if str(df.iloc[i,6])>=en:
        d_1.append(i)
        print(df.iloc[i])
        print("_____")
        i+=1
    else:
        i+=1
print(d_1)
```

```
[ ]
en=str(input("enter engine type : "))
i=0
d_1=[]
while i<8244:
    if str(df.iloc[i,6])>=en:
        d_1.append(i)
        print(df.iloc[i])
```



```
print("_____")  
_____  
    i+=1  
else:  
    i+=1  
print(d_1)
```

```
[]  
df
```

```
[]  
dw=str(input("enter drive wheel type : "))  
d_3=[]  
for t in d_2 :  
    if str(df.iloc[t,10])==dw:  
        d_3.append(t)  
        print(df.iloc[t])  
        print("_____")  
_____
```

```
[]  
co=str(input("enter category : "))  
d_4=[]  
for t in d_3 :  
    if str(df.iloc[t,3])==co:  
        d_4.append(t)  
        print(df.iloc[t])  
        print("_____")  
_____
```



```
[ ]
co= input("Gear box type(Automatic/Tiptronic/Manual/Variator) : ")
d_5=[]
print("\nTHESE are the cars witch fit you most : \n")
for t in d_4 :
    if str(df.iloc[t,8])==co:
        d_5.append(t)
        print(df.iloc[t])
        print("_____")
print(d_5)
```

5.Results and Key Findings :

Customized Suggestions:

Our system did a great job suggesting cars that matched what people wanted. By considering things like budget, preferences, and needs, it gave personalized recommendations that suited users well.

Better User Experience:

People found our system easy to use. They liked how simple it was to tell us what they wanted, and they appreciated getting suggestions that made sense for them.

Adapting to Different Places:

Our system was smart enough to think about the differences across India. It considered things like roads and fuel availability in different areas, making sure the suggestions were a good fit for each place.

Quick and Smart:

Our system worked fast! It handled a lot of car data and still gave good suggestions quickly. It showed how efficient and smart our recommendation system was.

Learning from Users:

Feedback from users was super helpful. It gave us ideas on how to make our system even better. Their input helped us improve the system to provide even more accurate and helpful suggestions.

Looking Ahead:

We found ways to make our system even cooler in the future. We want to add more car info, make the suggestions even more precise, and include the latest car models to keep our recommendations up to date.

Overall, our system did a great job suggesting cars that suited people's needs in India. It showed us ways to make choosing a car easier and more personalized, and we're excited to keep making it even better.

6.Conclusion :

Personalization and Precision:

- AutoMatch's strength lies in its ability to understand and adapt to individual preferences. The system continuously refines its suggestions based on user feedback, ensuring a high degree of personalization and precision in car recommendations.

Time and Effort Savings:

- The project significantly reduces the time and effort traditionally required for researching and selecting a car. Users can make well-informed decisions without the need for extensive manual searches, leading to a more efficient and satisfying experience.

Comprehensive Information Access:

- AutoMatch doesn't just provide suggestions but equips users with comprehensive information about each recommended car. From specifications to user reviews, the platform empowers users with the knowledge needed to make confident choices.

User-Friendly Interface:

- The user-friendly interface ensures a seamless and intuitive navigation experience. Whether you are a seasoned car enthusiast or a first-time buyer, AutoMatch caters to users of all backgrounds, making the process accessible and enjoyable.

Continuous Improvement:

The iterative feedback loop allows the system to continuously evolve and improve its understanding of user preferences. This commitment to refinement ensures that AutoMatch remains at the forefront of delivering accurate and relevant car recommendations.

6. Decision-Making Support:

Beyond recommendations, AutoMatch provides decision-making support by offering insights into long-term cost considerations, reliability ratings, and other factors crucial to making an informed choice.

7.References :

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